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Penile Geometry Self-Scanning Device Vibration, Haptic, and Tactile Guidance Integration

Field

This disclosure relates to self-operated anatomical geometry scanning devices and, more particularly, to the integration of vibration, haptic, and tactile guidance systems within a wearable or handheld penile geometry scanning apparatus.

Overview

In certain embodiments, the penile geometry self-scanning device further incorporates one or more vibration or haptic actuation elements. These elements are configured to provide tactile guidance, positional feedback, scan confirmation, perceptual signaling, or optional sensory engagement during geometry acquisition.

The device may be configured as a C-shaped or fully enclosed ring and is designed to be manually translated along the anatomy while capturing geometric measurements such as length, diameter, contour, or proportional relationships.

Actuator Technologies

The vibration or haptic subsystem may include one or more of the following actuator types, individually or in combination:

- Low-profile Linear Resonant Actuators (LRA)
- Miniature Eccentric Rotating Mass (ERM) vibration motors
- Piezoelectric vibration or surface actuators
- Localized haptic feedback elements integrated into the device housing

Actuator selection may prioritize low amplitude, fast response time, and minimal interference with optical or distance-based sensing subsystems.

Actuator Placement and Integration

One or more actuators may be positioned within the geometry scanning device, including but not limited to:

- Inner circumferential surfaces of the ring
- Discrete zones aligned with optical flow sensors, TOF sensors, or LiDAR sensors
- External touch points or grip regions for user feedback
- Mechanically isolated compartments to reduce vibration transfer into sensing elements

Tactile Guidance and Feedback Functions

Vibration or haptic output may be used to provide tactile cues, including but not limited to:

- Indication of correct initial positioning
- Guidance for consistent scanning speed or direction
- Feedback indicating adequate contact pressure or alignment
- Confirmation of scan initiation, progress, completion, or error states

Haptic cues may replace or supplement visual or auditory feedback.

Stimulus Characteristics

Haptic or vibration output may be modulated across one or more parameters, including:

- Frequency or resonance
- Amplitude or intensity
- Pulsed, continuous, or patterned waveforms
- Sequential or directional activation across multiple zones

Stimulus parameters may be fixed, adaptive, or user-configurable.

Sensor-Synchronized Operation

In certain embodiments, vibration or haptic output is synchronized with geometry acquisition sensors, including optical flow sensors, time-of-flight sensors, inertial measurement units, pressure sensors, or spoof-resistance sensors. This coordination supports scan consistency, artifact reduction, and improved repeatability.

Optional Sensory Engagement

In some embodiments, vibration or haptic actuation may optionally provide sensory or arousal-related engagement during the self-scan process, while remaining secondary to geometry acquisition and measurement accuracy.

Safety and Control

The device may include safety mechanisms such as intensity limits, time-based actuation constraints, automatic deactivation conditions, or user override controls to ensure safe and controlled operation during manual scanning.